

Technical Requirements Document
for
ESIF-HPC-4

Status: Draft v1.0

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1.0 Introduction

The Alliance for Energy Innovation (“The Alliance”), on behalf of the National Laboratory of the Rockies (NLR), has released a Request for Proposal (RFP) for the next-generation High Performance Computing (HPC) system to be installed at NLR’s Energy System Integration Facility (ESIF). This new system, which we refer to as ESIF-HPC-4, is expected to be delivered in mid-2027 and ready for production in 2028. The primary objective of the ESIF-HPC-4 procurement is to provide a computational system that will support the mission computing needs of the Department of Energy's Office of Critical Minerals and Energy Innovation (CMEI).

This document contains the technical requirements for ESIF-HPC-4. Each proposed solution in response to this document should clearly describe the role of the major lower-tier technology subcontractors and third-party software partners and the value added that these subcontractors provide.

These draft requirements describe specific technical requirements related to both the hardware and software capabilities of the desired system as well as application benchmark requirements. NLR aims to procure a highly productive, robust system that handles a diverse workload comprised of a mix of traditional modeling and simulations workloads as well as workloads in artificial intelligence (AI) and machine learning (ML).

The delivery timing provided in Section 1.3 represents current outlook and alignment of programmatic requirements and funding. “Alliance” reserves the right to revise the above schedule based upon their and/or DOE’s needs.

1.1 Guidance for Offerors

These requirements are structured with the objective of allowing offerors to propose a wide range of solutions, including an on-premise or off-premise system. Specific guidance for preparing responses is given below. Offerors may provide one or more proposals for the scenarios.

Offerors should respond to as many requirements and options as are applicable to the proposal and are practicable but should not feel obligated to respond to all. Offerors should mark all requirements, and label requirements not offered as Not Offered.

In Section 10, The Alliance requires maintenance for the first five years post-acceptance with options for maintenance in years 6 and 7. If offeror is bidding a service for which maintenance is included in the service price, describe the maintenance as requested in the requirements and indicate in the separate Pricing Sheet that the cost is included.

1.1.1 Guidance for On-Premise Proposals

NLR has a data center at the Energy Systems Integration Facility with power delivery capability of 10MW. The requirements section 9 provides information on the ESIF facility. Appendix B: Facilities section has additional information on the location and capabilities.

1.1.2 Guidance for Off-Premise Proposals

NLR intends for the language in these Technical Requirements to be inclusive of off-premises system proposals. If proposing an off-premises solution, the requirements of Section 9: Facilities do not apply and require no response.

NLR expects as much detail in the System Description regarding processors, nodes, interconnect, and storage as would be provided in an On-Premises proposal. As part of the Software response, indicate the level of service being proposed and how software will be provided (e.g., base operating system, programming environment, workload manager, storage) to NLR to deploy on the system.

1.2 Requirements Definitions

Requirements in this document have priority designations, which are defined as follows:

- Mandatory Requirements (MR) are features that are essential to NLR's requirements, and an Offeror must satisfactorily propose all Mandatory Requirements in order to have its proposal considered responsive.
- Target requirements designated as (TR-1, TR-2, or TR-3)

Target Requirements (designated as TR-1, TR-2, or TR-3) are features, components, performance characteristics, or other properties that are important to NLR. Target Requirements are prioritized by dash number, where TR-1 is most desirable to NLR, while TR-2 is more desirable than TR-3. Target Requirement responses will be evaluated as part of the proposal evaluation process.

- Technical Option Requirements (designated as TO-1, TO-2, or TO-3)

Technical Option Requirements (designated TO-1, TO-2, or TO-3) are features, components, performance characteristics, or upgrades that are important to NLR. Technical Options add value to a proposal. Technical Options are prioritized by dash number. TO-1 is most desirable to NLR, while TO-2 is more desirable than TO-3. Technical Option responses will be evaluated as part of the proposal evaluation process; however, NLR may or may not elect to include Technical Options in the resulting subcontract(s). Each proposed TO should appear as a separately identifiable item in an Offeror's proposal response.

1.3 Schedule

The following is the tentative schedule for the HPC-4 system acquisition:

Table1

Request for Proposals release	Q4 FY26
Offeror questions due	Q4 FY26
Proposals due	Q1 FY27
Early access hardware delivery	FY27
Full system delivery	Q1 FY28
System acceptance	Q2 FY28

1.4 Collaboration and Partnership

The NLR Energy System Integration Facility (ESIF) is collaborative research hub for future-ready, secure, reliable, and affordable energy technologies. As the home of NLR’s high-performance computing systems, the ESIF HPC Data Center serves as a living laboratory where advanced computing, AI workloads, and physical energy systems converge. Within this environment, NLR’s HPC systems are tightly integrated with experimental facilities to enable rapid materials screening, digital twins, hardware-in-the-loop experimentation, and real-time experimental steering. The HPC platform functions as a computational backbone for close-loop workflows in which modeling, simulation, and machine learning operate in concert with laboratory instrumentation – accelerating translation from discovery to scale-up and commercialization.

Areas of particular interest for collaboration include partnerships fostering innovation across the full stack from advanced semiconductor technologies and system architectures to facility integration and grid-responsive operations. NLR is interested in partnerships that explore new approaches to performance-per-watt optimization, composable and heterogeneous architectures, advanced interconnect and storage technologies, and hardware–software co-design under real operational energy constraints. Additional areas include digital twin frameworks and workload management strategies that enhance data center flexibility and grid interactivity, as well as AI-driven autonomous experimentation enabled by tightly coupled compute and experimental systems. These collaborations may span core HPC platforms, edge-to-core integration, real-time data pipelines, and emerging multi-modal computing paradigms that accelerate applied energy research and deployment.

Looking beyond the installation of new computational capabilities, NLR welcomes proposals that include substantial future collaboration and partnership with the successful Offeror. These could take the form of joint institutes, a new center of excellence, co-development of AI-optimized and power-efficient computing architecture, product testing and validation within a grid-connected research environment, or coordinated product announcements. Proposed opportunities should be

mutually beneficial to both organizations and be consistent with NLR's leadership role in applied energy research and AI-enabled energy innovation.

2.0 Workload Overview

NLR's systems have historically supported a heterogeneous mix of traditional modeling and simulation workloads, as well as high-throughput ensembles of smaller jobs. AI/ML-driven workloads are increasingly becoming a larger percentage of compute, as are hybrid and data intensive workloads. Specifically:

- **Artificial Intelligence:** AI and machine learning (AI/ML) is a growing workload on NLR systems. As researchers and application developers explore and integrate AI/ML technologies into their workflows, a broad range of use cases is anticipated. These workloads are comprised primarily of training and inference jobs ranging from single-node to multi-node scale requiring high memory bandwidth and intra-node GPU-GPU communication. Large language models (LLMs) and generative AI workloads are also experiencing growth. Many workloads take the form of workflows that couple AI/ML with traditional simulation codes, requiring double-precision data types and fast data movement between CPUs, GPUs and storage. These AI/ML-powered workflows will possibly require an optimized storage layer that can handle high I/O, i.e., frequent reads of large number of training files and model artifact writes.
- **Modeling and Simulation:** Characterized by large-scale, tightly coupled jobs spanning hundreds of nodes with both strong and weak scaling requirements, and requiring high-precision data types (i.e. FP64). The largest runs scale to ~50% of the system, though the majority typically consume 5-10% of compute resources per job. In addition, this category includes high throughput ensembles of smaller jobs using fewer than 100 nodes, but are often single node only. Simulation codes are increasingly migrating to accelerated paradigms, and many require scaling to multiple GPU nodes. A minority of the modeling and simulation workloads will continue to require CPU-only nodes. The interconnect needs to provide sufficient support for all modeling and simulation workloads, provide application scalability, and balance node-level computation performance.
- **Hybrid and Data Intensive:** Includes digital twins, distributed data analytics, visualization, experimental interoperability, and edge compute capabilities, often requiring container-based deployments or orchestrated workflows. Some use cases require real-time on-demand access to compute resources rather than the traditional HPC queue, to enable operational digital twin research, AI-controlled experimental facilities, and hardware-in-the-loop experiments.

3.0 System Architecture Requirements

The system requirements have been split into four zones: high performance compute zone, access zone, storage zone, and management zone. These zones are based on NIST SP 800-223 and are a logical segmentation of the HPC system.

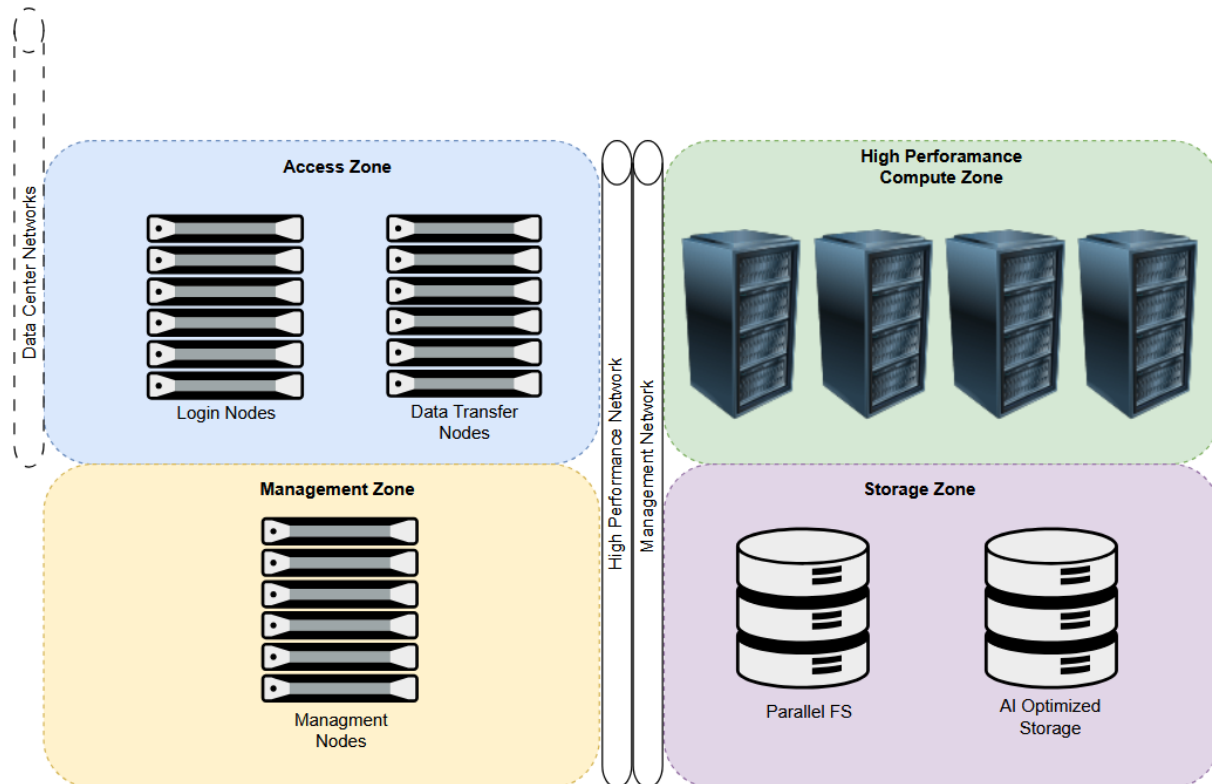


Figure1

3.1 System Description

The Offeror shall provide a detailed architectural description of the proposed compute system, including text, diagrams, tables, etc. as needed. Provide descriptions of each processor type, node type (which and how many processors, connectivity between processors, memory, NICs, node-local storage, if any), blade/chassis type (if applicable), rack/cabinet and interconnect. The offeror should include quantities and define any minimum scalable unit sizing to maintain optimal performance and productivity across the system.

[MR] Provide a high-level overview of the system by describing the items below, if applicable:

- Component architecture – details of all processor(s), memory technologies, storage technologies (if used on the compute nodes), network interconnect(s) and any other applicable components.
- Compute node architecture(s) – details of how components are combined into the node architecture(s). Details should include bandwidth and latency specifications (or projections) between components. Details should be provided for each compute node type.

- Board and/or blade architecture(s) – details of how the node architecture(s) is integrated at the board and/or blade level. Details should include all inter-node and inter-board/blade communication paths and any additional board/blade level components.
- Rack and/or cabinet architecture(s) – details of how board and/or blades are organized and integrated into racks and/or cabinets. Details should include all inter rack/cabinet communication paths and any additional rack/cabinet level components.
- Interconnect - details of the system's high speed network topology and connectivity across all system components (compute nodes, login nodes, management system).
- System architecture – details of how rack or cabinets are combined to produce system architecture, including the high-speed interconnects and network topologies (if multiple).
- Management node(s) - details of hardware to support management and services to operate the ESIF-HPC-4 system. Describe the management node types (e.g., admin, leader, scheduler/resource manager, fabric manager). Multiple node types may be needed to optimize for different uses. Management nodes should be accessible even if the compute system is unavailable.
- Access zone nodes- details of hardware to support user access and potentially user-driven workflow activities. A pool of access zone nodes will be needed to address the requirements described in Section 7. The access zone will collectively support 400 interactive users, 60 data analysis front-end applications, 150 integrated development environments and 100 simultaneous compilations of software of equivalent complexity to the latest GNU Compiler Suite. The access zone should be accessible even if the high performance compute zone is unavailable.

3.2 Requested Options

The Offeror will provide Technical Options to enable NLR to adjust various components of the system. Only provide descriptions of the options in this document; do not include prices. Prices should be made available separately in the Price Breakdown Table.

Technical Options may include architectural choices such as additional racks, changes to memory capacity or interconnect topology. The Offeror should describe which options remain viable after deployment (e.g. adding racks of CPU-only nodes, adding AI nodes, incremental storage adds) and for how long after deployment.

3.2.1 Hardware Options

3.2.1.1 [TO-1] Scale the System Size

The Offeror should provide an option in the Price Breakdown Table to increase and decrease the compute partition. Provide options separately for each node type proposed. Describe here the scalable unit (e.g., rack, N racks with CDU, row). The Offeror should identify any thresholds requiring increased component infrastructure (e.g., extra switches, extra cables), any technical challenges foreseen with respect to scaling and any other production issues.

3.2.1.2 [TO-1] Scale Processor Memory

For each processor type, the Offeror should provide an option to increase and decrease, if possible, the memory capacity. Describe the scaling units. The price should be per scaling unit (e.g., rack, N racks).

3.2.1.3 [TO-2] Scale the Interconnect

The Offeror should provide options to increase and decrease the interconnect bandwidth. This option may include additional NICs per node, additional switches, additional cables, etc. Describe the expandability of the interconnect and identify any thresholds requiring increased component infrastructure (e.g., extra switches, extra cables), any technical challenges foreseen with respect to scaling and any other production issues.

3.2.1.4 [TO-1] Additional Maintenance Periods

The System Price should include five years maintenance. The Offeror should provide separate options for year 6 and year 7 maintenance and support for the system. If the Offeror is a cloud provider, provide the price to continue the service for each year.

3.2.1.5 [TO-3] Mid-Life Upgrades

The Offeror should describe and separately price any future options for adding node types not available during the proposal period to the proposed ESIF-HPC-4 system over its five-year lifetime.

3.2.1.6 [TO-3] Data, Analysis and Visualization Nodes

In order to provide additional visualization/rendering capabilities, the Offeror may propose a price for a scalable unit (e.g., rack, N racks with CDU) of data, analysis and visualization/rendering nodes that connect to the high speed interconnect. Describe the node architecture including processor architecture, number of processors, memory, connectivity within the node, connectivity to the high

speed interconnect, and node-local storage, if any. The Offeror may choose to provide multiple options and prices for different architectures.

3.2.1.7 [TO-3] Node-Local Storage

For each node type, the Offeror should provide an option to add NVMe drives with a target size of at least triple the node memory capacity. Describe the NVMe performance characteristics and system software requirements.

3.2.1.8 [TO-3] Advanced Architectures Option

NLR is interested in optional advanced or novel architectures that can accelerate AI, data-driven analytics, modeling and simulation, and other emerging scientific workflows. These may include novel AI/ML acceleration technologies, heterogeneous or domain-specific processors, dataflow or reconfigurable architectures, neuromorphic approaches, or other innovative designs capable of integrating into the HPC ecosystem.

Proposed options should describe how the architecture would enhance performance, scalability, and workflow efficiency for workloads characterized by high-throughput simulation, AI-driven model development and tight integration with experimental facilities. The option may be delivered with the system or later and may consist of scalable units defined by the Offeror. Multiple options may be proposed and should be described and priced separately. Offerors may highlight architectural features that provide clear benefits for integrated AI/HPC workflows, rapid iteration cycles, real-time or streaming data pipelines, or acceleration of domain-specific numerical kernels.

For each proposed Advanced Architectures option, the Offeror shall provide:

- An architectural description: an overall architectural diagram showing hardware, interconnect(s), compilation infrastructure, and I/O subsystems; and confirmation that the partition can mount NLR's file system(s) at the required performance.
- A software environment description: an overview of the software stack, including libraries, SDKs, supported frameworks, usability/programmability characteristics, and all associated licensing and support terms.
- A performance characterization: performance results (measured or projected) for relevant benchmarks listed in Section 8, with the benchmark version, submission status, and any rule modifications clearly specified.

3.2.3 Early Access Technology

[TO-3] The Offeror should describe options to provide NLR with early access hardware that enables hardware and software testing prior to delivery of the production system. Early-access systems may use processor generations one or two generations older than the processors proposed for the production system (i.e., *N-1* or *N-2*). Early access to other key technologies, such as interconnects, memory, or accelerators, is also encouraged. If multiple early-access configurations are offered, each should be described and priced separately.

3.2.4 Alternatives to Stand-Alone Test and Development System

NLR seeks alternatives to a traditional stand-alone Test and Development System (TDS). Because such systems are often difficult to maintain in lockstep with the production system, NLR is interested in approaches that leverage the early-access resources described in Section 2.2.3 or that make use of the delivered system itself.

Prior to substantial system delivery, NLR is interested in mechanisms that provide developers and user-support staff with early system exposure to accelerate application readiness and reduce time-to-production. These mechanisms may include access to pre-delivery hardware or early access hardware as described in Section 3.2.3 that mirrors the configuration of the proposed system; or off-premises resources comparable to the proposed system for application porting, training, or workflow development.

After substantial delivery, NLR is interested in how the production system can be partitioned or segmented to support development activities, such as testing operating system updates, firmware revisions, system software changes, or application-stack upgrades, without impacting broader system availability.

3.2.4.1 [TO-1] Pre-delivery options for Test and Development

The Offeror shall describe available options for providing pre-delivery access or resources suitable for early development and training.

3.2.4.2 [TO-1] Post-delivery options for Test and Development

The Offeror shall describe how the delivered system can be partitioned, isolated, or otherwise configured to support testing and development while maintaining production operations. Describe any additional hardware, software, or architectural features that enable these test and development workflows.

3.2.5 Off-premise Options

NLR is interested in augmenting on-premises compute capacity with off-premise, consumption-based (cloud) resources to support bursting or time-sensitive workloads. Cloud resources may be used for overflow demand, urgent mission-driven work, or workloads that require hardware capabilities not available on-premises.

The Offeror should provide options for providing a guaranteed pool of compute capacity (e.g., node-hours or equivalent metrics) on a monthly or quarterly basis for the life of the production system. Proposed solutions should include mechanisms for securely accessing cloud resources, expected performance characteristics, and any data-movement considerations.

For Off-premise options, the Offeror shall provide:

3.2.5.1 [TO-1] Security

All proposed cloud resources must be FedRAMP-authorized at a level suitable for NLR workloads.

3.2.5.2 [TO-2] Capacity and pricing

Descriptions of guaranteed capacity options and associated pricing models.

3.2.5.3 [TO-3] Integration

Any mechanisms for integrating or extending on-prem workflows to cloud environments (e.g., bursting, federated schedulers, or containerized execution).

3.2.5.4 [TO-3] Other considerations

Any constraints, data-transfer requirements, or architectural considerations relevant to the proposed solution.

4.0 High-Performance Compute Zone

The High-Performance Compute Zone contains the compute nodes and the high-performance interconnect which connects the compute nodes to the other zones. NLR's anticipated workload will be diverse, and serve AI, high-throughput, high-performance, and data-centric computing models. NLR anticipates the majority of the compute cycles will be provided by accelerated nodes.

4.1 Compute Nodes

4.1.1 Compute node architecture

[TR-1] Offeror shall provide a detailed description of each proposed compute node type, including:

- Processor or accelerator architecture
- Memory types, configuration, and maximum capacity
- Network interface type(s) and link topolog
- Per-network-interface bandwidth and latency, and the effective per-node network bandwidth
- Per-node peak computational performance and sustained memory bandwidth
- High-bandwidth memory (HBM) capacity provided on accelerated node types, per accelerator. It is highly desirable that the system provide at minimum double the HBM capacity per accelerator than Kestrel

4.1.2 Compute node ratio

[TR-2] The Offeror shall describe the compute node architecture ratio of the proposed system. A CPU-only (Arm or X86_64) partition shall represent no more than 10% of the system's total peak FLOP capacity.

4.1.3 Compute Processor Energy Efficiency

[TR-2] It is highly desirable that the system support operation at the optimal point on the performance-per-watt curve during computationally intense phases. The Offeror shall describe any user-controllable power or frequency features available on compute processors, accelerators, NICs, and switches.

4.2 High Speed Interconnect

4.2.1 Network Description

[TR-1] The Offeror shall describe the high-speed network including:

- A high-level description of topology and routing protocol
- The per-node and per-link network bandwidth and latency characteristics
- The number and type of high-speed network interfaces on each compute node
- Any hardware-accelerated communication features, such as GPUDirect, GPUDirect RDMA, or equivalent technologies
- Any network-level congestion management features relevant for accelerated workloads.

4.2.2 Resilience

[TR-1] The Offeror shall describe the network's resilience features, including link- and switch-failure scenarios, the number of failures tolerated while maintaining connectivity, and the expected impact on performance as components fail.

4.2.3 Management

[TR-1] The Offeror shall describe the out-of-band management network, including supported security mechanisms, routing and segmentation capabilities.

4.2.4 Quality of Service

[TR-2] The Offeror shall describe the interconnect QoS mechanisms, including congestion control, traffic classes/virtual channels, and prioritization options. The Offeror should describe any recommended QoS configurations for typical HPC workloads (messaging, collective operations, and storage IO) and any features that support complex or multi-tenant workflows.

4.2.5 Compute Network Integration

[TR-2] The Offeror shall describe the capability of the high-speed network to integrate storage systems, specialized nodes, or other vendor-supplied components into the fabric, including any limitations on interoperability or mixed-vendor configurations.

4.2.6 Protocol Support

[TR-1] The Offeror shall describe protocol support for both the high-speed and management networks, including:

- Supporting IP networking capabilities (IPv4/IPv6),
- Capabilities for traffic control and ACLs on management interfaces,
- Mechanisms for path selection, routing, and load balancing where applicable.

4.2.7 CXL

[TR-3] If CXL is offered, the Offeror shall describe supported CXL versions and capabilities for any CPU, accelerator, NIC, or storage components that utilize CXL.

4.2.8 MPI

[TR-1] The system shall provide MPI libraries supporting MPI 4.0 or higher, with accelerator-aware MPI where supported by the accelerator vendor. The Offeror shall describe any deviations from or extensions to the MPI standard.

4.2.9 PMI

[TR-1] The system shall support the Process Management Interface (PMI). The Offeror shall describe the supported PMI versions and any relevant integrations.

4.2.10 Other Communication Libraries

[TR-3] The Offeror shall describe any additional communication libraries provided (e.g., SHMEM, UCX-based PGAS libraries, etc).

4.2.11 Optimized Collective Libraries

[TR-3] The Offeror shall describe any optimized collective-communication libraries provided (e.g. NCCL, RCCL, vendor-specific collective libraries) and any noted performance characteristics or limitations.

5.0 Access Zone

The Access Zone provides the primary user and administrative entry point to the system. It must support secure authentication, data ingress/egress, interactive workflows, and system management functions while maintaining compliance with DOE's cybersecurity requirements. The Login and DAV Nodes should provide an architecture and programming environment compatible with the High-Performance Compute Nodes. NLR will provide dual 100G or dual 25G Ethernet switch connections for every system in the Access Zone to connect to the NLR data center network or Science DMZ. For user accessed resources like login nodes and data transfer nodes, 100G Ethernet is preferred.

5.1 Access Zone Architecture

5.1.1 Architecture

[TR-1] The Offeror shall describe how the proposed Access Zone architecture provides login, data transfer, and analysis/visualization services, supporting 400 interactive users, 60 data analysis front-end applications, 150 integrated development environments and 100 simultaneous compilations of software of equivalent complexity to the latest GNU Compiler Suite.

5.1.2 Access Zone compatibility

5.1.2.1 [TR-1] Component Interoperability

The Offeror shall describe how Access Zone components interoperate with the Compute Zone, including compatibility with the system software stack, storage systems, and cluster management, without requiring workflow modifications from users.

5.1.2.2 [TR-2] Login and DAV nodes

Login and DAV nodes are listed separately by function in this document. Describe if the two node types can be provided by the same physical node type.

5.2 Access Nodes

5.2.1 Login Nodes

The system should support interactive access for command-line session, pre- and post-processing, and development workflows like Jupyter or VSCode.

5.2.1.1 [TR-1] Login node architecture

The Offeror shall describe the login node architecture that will meet these requirements and provide sufficient compute, memory and I/O for concurrent multi-user workloads.

5.2.1.2 [TR-2] Login node connectivity

Login nodes should connect to the Ethernet data center network with dual connections preferred.

5.2.2 Data, Analysis, and Visualization Nodes (DAV)

NLR uses Data, Analysis and Visualization nodes for pre and post processing as well as other interactive graphical work. Typically, these are deployed with a virtual desktop provided by NLR such as FastX.

5.2.2.1 [TR-1] DAV Nodes

The system should support interactive analysis and remote visualization of simulation and experimental data. The Offeror shall describe a set of at least 15 DAV nodes that incorporate accelerators, memory capacity and software

5.2.2.2 [TR-1] OpenGL

Describe the support for OpenGL to enable remote visualization and interactive exploration of large datasets.

5.2.3 Data Transfer Nodes

Data Transfer nodes will be used for secure, high-throughput data transfer between the system's file system and external networks. Due to the sensitive location of these nodes, a fully compatible environment with the cluster is not necessary in order that these systems can be patched more aggressively than the rest of the cluster.

[TR-1] The Offeror shall describe the architecture of at least four data transfer nodes providing high-speed external network connectivity.

5.2.4 External Connectivity

[TR-1] The Offeror shall provide a high-bandwidth and resilient solution for connectivity between the high-speed compute fabric and the ESIF data center network. The solution shall support simultaneous transfers and provide aggregate throughput commensurate with the expected system IO and data-movement workloads.

5.2.5 Off-premise External Connectivity

[TR-3] If any system components or resources are deployed off-premises, the Offeror shall describe external connectivity between NLR and the off-premises resource. Any Offeror-provided fiber connection shall terminate in the ESIF Data Center.

5.3 User Environment

5.3.1 User Environment Software Management

5.3.1.1 [TR-3] Module Framework

The Offeror shall describe the mechanisms by which the provided user software libraries (mathematical, I/O, and AI/ML; 5.2.2) can be installed for each provided toolchain supported on ESIF-HPC-4. Note how multiple versions of packages are managed and made accessible.

5.3.1.2 [TR-3] Container Build and Execution Capabilities

Describe support for unprivileged users to build and run containerized applications while leveraging provided system hardware and software features. Note whether the proposed container technologies support, or comply with, Open Container Initiative (OCI) standards.

5.3.1.3 [TR-3] Containerized Programming Toolchain

To support portable development workflows, NLR expects ESIF-HPC-4 to supply a containerized programming toolchain (i.e., compiler(s) and MPI) that mirrors the host environment and can be used both on and off ESIF-HPC-4. Please describe the offering and its limitations.

5.3.2 Scientific Programming and Development

5.3.2.1 Compilers and Programming Languages

ESIF-HPC-4 must support a broad set of compiled and interpreted languages. The Offeror shall describe how:

5.3.2.1.1 [TR-3] The latest ISO standards for C and C++ are supported, noting any partial or missing features.

5.3.2.1.2 [TR-3] The Offeror shall describe how more than one programming toolchain to program all compute node types.

5.3.2.1.3 [TR-2] For compilers requiring licenses, provide separate pricing for at least 10 seats and temporary licensing sufficient for benchmarking and acceptance testing.

5.3.2.2 CPU threading

5.3.2.2.1 [TR-1] The system must support OpenMP threading. The Offeror shall describe any limitations in OpenMP threading support, if applicable.

5.3.2.3 Accelerator programming

5.3.2.3.1 [TR-1] The Offeror shall describe support for accelerator-based programming platforms and/or APIs, including any associated tools for conversion between different stacks (if applicable).

5.3.2.3.2 [TR-1] Describe available programming models for directive-based accelerator offloading.

5.3.2.4 AI/ML Libraries and Frameworks

[TR-3] The Offeror shall describe any optimized frameworks and/or libraries available to maximize performance for AI/ML tasks such as distributed training, inference, and hyperparameter tuning. Describe how any provided frameworks and/or libraries integrate with and utilize the AI Optimized Storage (section 6.2).

5.3.3 Mathematical and I/O Libraries

5.3.3.1 Optimized Mathematical Libraries

[TR-1] The Offeror will provide optimized BLAS, LAPACK, SCALAPACK, and FFT libraries for each compute node type and supported precisions. Please describe the offering and its limitations.

5.3.3.2 Optimized I/O Libraries

[TR-1] The Offeror shall describe support for optimized I/O libraries (e.g., NETCDF, p-NETCDF, HDF5) on each compute node type, including any parallel I/O capabilities and limitations.

5.4 Performance Analysis Tools and Debuggers

NLR currently uses several third-party performance analysis and debugging tools (e.g., Linaero Forge) to support existing and emerging HPC workflows. ESIF-HPC-4 must be able to support these types of tools for ongoing code maintenance and support by NLR staff.

5.4.1 Performance Profiling

5.4.1.1 [TR-2] Profiling Tools

The Offeror shall describe how the system provides performance profiling and tracing tools that operate across all supported programming models (e.g. MPI, OpenMP, accelerator programming) and compute node types.

5.4.1.2 [TR-1] Hardware Counters and Multi-node profiling

The Offeror shall describe access to hardware counters and support for multi-node profiling.

5.4.1.3 [TR-2] Storage and I/O Performance Profiling

The Offeror shall describe available tools for profiling storage and I/O performance, including collection of throughput and metadata operation metrics relevant to user workflows.

5.4.2 Debugging

[TR-2] The Offeror shall describe the provided debugging environment, including graphical and command-line interfaces, and how it supports debugging of parallel applications across multiple nodes, threads, and accelerators.

6.0 Storage Zone

The Storage Zone must support a wide range of computational workloads, each placing different demands on the storage environment. Traditional modeling and simulation require a high-bandwidth, write-optimized parallel file system capable of supporting large data outputs, checkpoint/restart operations, and mixed I/O patterns ranging from small metadata-intensive files to large streaming transfers. At the same time, emerging AI/ML workflows introduce access patterns that differ substantially from conventional HPC usage. These workloads often rely on small, random reads and may require higher IOPS and lower latency characteristics. To accommodate these needs, NLR is willing to consider an additional storage tier optimized for AI and data-driven workflows, capable of serving as a shared read resource across many nodes within a job.

Both traditional and AI-focused storage use cases must provide consistent performance at varying scales, from single node jobs to workloads utilizing a substantial portion of the system, ensuring that users can rely on predictable behavior across ESIF-HPC-4's diverse computational workloads. All required file systems (e.g. Scratch, AI-Optimized Storage, Projects and Home) must be mountable across the Access and Compute Zones in a consistent manner.

6.1 Parallel File System

The system must include a high-performance parallel file system supporting modeling and simulation workloads with high bandwidth and mixed I/O patterns.

6.1.1 Architecture

[TR-1] The Offeror shall describe the architecture of the proposed parallel file system, including key hardware and software components, network connectivity, supported interfaces (e.g. POSIX), and any required client-side modules.

6.1.2 Capacity and Performance

[TR-1] The parallel file system should exceed a capacity of 50PB and a sustained read and write performance of 500Gb/s. The Offeror shall describe how the system meets or exceeds these minimum requirements, and how performance scales from single-node jobs to system-level workloads.

6.1.3 Resilience and Failure Behavior

[TR-1] The Offeror shall describe data-protection mechanisms, expected rebuild behavior, and performance impacts while operating in degraded or recovery modes.

6.1.4 Management, Monitoring and Diagnostics

[TR-1] The Offeror shall describe the administrative tools provided for performance monitoring, diagnostic analysis, quota enforcement, and usage reporting.

6.1.5 Expandability

[TR-1] The Offeror shall describe how additional storage or metadata capacity may be added after deployment, including any performance implications or downtime requirements.

6.1.6 Security Controls

[TR-1] The Offeror shall describe security features, including at-rest and in-transit encryption and any capabilities to restrict visibility of sensitive or project-scoped data.

6.2 AI-Optimized Storage

To support emerging AI, ML, and data-driven workloads, the system may include a second storage tier optimized for high-IOPS, low-latency access and shared dataset availability.

6.2.1 Architecture

[TR-1] The Offeror shall describe the architecture of the proposed AI-optimized storage tier, including hardware/software components, supported interfaces (e.g. POSIX, S3/object, APIs), and any client-side software requirements.

6.2.2 Capacity and Performance

[TR-1] The AI-optimized storage tier should exceed a capacity of 50PB and a sustained read and write performance of 100 GB/s. The Offeror shall describe how the system meets or exceeds these minimum requirements, and how it serves high-IOPS, small-file and random read access patterns characteristic of AI/ML workloads.

6.2.3 Data Sharing and Protection

[TR-1] The Offeror shall describe capabilities for snapshots, project level permissions, and isolation mechanisms preventing unauthorized access or data leakage across project boundaries.

6.2.4 Policy, Lifecycle and Monitoring Tools

[TR-1] The Offeror shall describe usage reporting, quota enforcement options, and any policy engine for automated data lifecycle operations, as well as monitoring and diagnostic capabilities.

6.2.5 Expandability

[TR-1] The Offeror shall describe how capacity and metadata services may be expanded over time, along with expected impacts on availability and performance.

6.2.6 Security Controls

[TR-1] The Offeror shall describe security capabilities relevant to collaborative and AI storage environments, including encryption, access controls, and audit/logging mechanisms.

7.0 Management Zone

7.1 System Management Architecture

The Management Zone includes the tools, networks, services, and security mechanism required to configure, operate, monitor, and maintain the ESIF-HPC-4 system. Access to the Management Zone shall be restricted to authorized administrative personnel.

7.1.1 Management Infrastructure

[TR-1] The Offeror shall describe the overall management architecture, including service nodes, management servers, management storage, and all software components supporting management operations.

7.1.2 Configuration and Cluster Management

[TR-1] The Offeror shall describe provided configuration management capability, either built-in or integrable with existing tools, and how system configuration consistency is maintained across upgrades.

7.1.3 Management Network

[TR-1] The Offeror shall describe the management network topology, how management traffic is isolated from user traffic, and whether a dedicated out-of-band network is included.

7.1.4 Remote Administration

[TR-2] The Offeror shall describe remote console and power control capability (e.g. IPMI, Redfish)

7.1.5 High-Availability and Upgrade Strategy

[TR-2] The Offeror shall describe how management services remain operational during upgrades or maintenance, and which operations require planned downtime.

7.2 Operating System and Workload Management

7.2.1 Base Operating System

[TR-1] The Offeror shall describe the Linux distribution used throughout the system, including licensing model, update cadence and kernel customization (if any).

7.2.2 OS Upgrade and Patching

[TR-1] The Offeror shall describe how OS upgrades, including security patches, can be performed while minimizing downtime and disruption to running workloads.

7.2.3 Job Scheduler and Resource Manager

7.2.3.1 [TR-1] Job Scheduler and Resource Manager

The Offeror shall describe the provided job scheduler and resource manager. If Slurm is not offered, the Offeror shall describe functional equivalence and architecture-level advantages for ESIF-HPC-4.

7.2.3.2 [TR-2] Shared Scheduling

The Offeror shall describe how the scheduler supports multiple simultaneous jobs and users on a single node where relevant.

7.3 System Monitoring

NLR will be active in the design, implementation and usage of cluster monitoring infrastructure. Data collected may be exported from the system for use as research data and may be linked with power and energy usage and jobs data.

7.3.1 Monitoring Capabilities

[TR-1] The Offeror shall describe provided monitoring tools capable of collecting system health, hardware status, environmental metrics, and performance information from compute, management, storage, network components, as well as available interfaces for integrating facility-level telemetry (e.g. CDU temperatures, flow rates, or power data) when exposed through standard protocols.

7.3.2 Sensors and Telemetry

[TR-1] The Offeror shall describe available sensors, supported telemetry protocols (e.g. Redfish, IPMI, SNMP, Modbus, OS-level monitoring), and maximum supported polling frequencies.

7.3.3 Data Export and APIs

[TR-2] The Offeror shall describe APIs or other mechanisms allowing NLR to export telemetry data to external systems (e.g. Prometheus, Grafana, research workflows).

7.3.4 Logging Infrastructure

[TR-2] The Offeror shall describe how system logs are aggregated and retained, with capacity to store at least seven days of cluster-wide logs without impacting system operation.

7.4 Power and Energy

7.4.1 Power and Energy Measurement Interfaces

[TR-2] The Offeror shall describe system- or rack-level interfaces capable of reporting node, CPU, GPU, memory, and facility-level power and energy metrics.

7.4.2 Power and Energy Control Features

[TR-3] If available, the Offeror shall describe optional control interfaces such as system-level power caps or cooling-related telemetry*(e.g. water temperature, flow).

7.5 Security

ESIF-HPC-4 will be a Federally owned system and NLR will be required to operate it in a secure manner. NIST publication 800 53 offers guidance on the operating of Federal systems and is one of the security frameworks NLR basis its security policy. Header designations below are taken from the NIST 800 53 documentation.

7.5.1 Preventing Program Execution

[TR-1] CM-7(2): Prevent Program Execution: Describe how offeror provided software and kernel drivers can be operated with standard Linux security architecture tools like selinux.

7.5.2 Inventory

[TR-1] CM-8(1): Updates During Installations / Removals: Upon delivery, updates and upgrades inventory will be provided or updated.

7.5.3 Backup and Restoration

[TR-1] CP-9 System Backup; CP-9(1): Testing for Reliability and Integrity; CP-9(2): Test Restoration Using Sampling; CP-10: System Recovery and Reconstitution; CP-10(2): Transaction Recovery; CP-10(4): Restore Within Time Period: Identify critical system level software, license, applications, middleware and configurations. Describe the process for backing up and restoring.

7.5.4 Replaceable Media

[TR-1] MP-6 Media Sanitization: Offeror shall describe the process of replacing media. NLR may return used media if properly sanitized, otherwise it will be destroyed.

7.5.5 Security Notification and Remediation

[TR-1] SI-2 Flaw Remediation; SI-7 Security Alerts, Advisories, and Directives: The Offeror shall describe its vulnerability handling and patch management process for vendor-supplied software and firmware, including expected timelines from confirmed availability for:

- Non-critical vulnerabilities
- Vulnerabilities listed in the CISA Known Exploited Vulnerabilities (KEV) catalog
- Common Vulnerabilities and Exposures (CVEs) with a score of Critical or High (per the current CVSS version of CVSS by NVD).

7.5.6 Cryptographic Protection and Remote access

[TR-1] SC-12: Cryptographic Key Establishment and Management; SC-13: Cryptographic Protection; AC-17(2): Protection of Confidentiality and Integrity Using Encryption: The system will use encrypted protocols for remote and web logins, including switches and BMC.

7.5.7 Monitoring and Endpoint Protection

[TR-1] SI-4(1): System-wide Intrusion Detection System; SI-4(2): Automated Tools and Mechanisms for Real-time Analysis; SI-4(3): Automated Tool and Mechanism Integration; NLR will install EDR (Endpoint Detection and Response) agents. Describe any known interaction issues with Offeror software from EDR tools.

7.5.8 Baselines

[TR-2] CM-2: Baseline Configuration: Describe the recommended secure baseline configuration for the system and how it is validated and documented. The Offeror shall describe methods to assess configuration drift and to support auditability of the system's current security-relevant configuration state.

7.5.9 Supply Chain Risk Management

[TR-1] SR-08, SR-10; SR-11; SR-12; Packaging of materials should be done in a way to easily detect tampering. Anti-counterfeit measures should be implemented. Safe disposal procedures should be followed to prevent the leaking of data and cryptographic keys. NLR should have early notification of compromise or potential compromises to the supply chain that could adversely affect the system or components of the system.

8.0 Benchmarks

Benchmark performance is a critical component of evaluating the proposed ESIF-HPC-4 system architecture. NLR will assess how well real scientific applications, data-intensive workflows, and key microbenchmarks perform on the offered solution. The Offeror is required to provide benchmark results for the application, microbenchmark, and I/O test defined in Tables 2-4, using the instructions and input configurations provided in the ESIF-HPC-4 benchmark repository (<https://github.com/NatLabRockies/ESIFHPC4/>).

NLR provides three categories for benchmark runs: baseline, ported, and optimized; to capture both architectural suitability and potential performance improvements achievable with reasonable development effort. These categories are defined to ensure comparability across proposals and to reflect realistic expectation for system readiness and application support at delivery.

The Offeror shall return benchmark results using the ESIF-HPC-4 Benchmark Results spreadsheet and shall include a full benchmark report in the Technical Volume Appendix. The Offeror is responsible for providing temporary licenses for any compilers or tools used to generate benchmark results that are not part of the delivered system.

An application throughput improvement metric derived from a subset of application benchmarks is used to evaluate system-level capability relative to the current Kestrel system. A detailed description of the application throughput improvement metric is provided in Appendix A. The application throughput improvement benchmark configurations represent realistic NLR workflow sizes and must demonstrate improvement over the provided reference performance results.

NLR encourages the Offeror to use best practice methods to obtain reliable benchmark results within the allowed modification categories. The Offeror shall adhere to benchmark-specific instructions and constraints documented in each benchmark's README.md file in the ESIF-HPC-4 repository.

8.1 Benchmark Categories

We define the following general rules for baseline, ported, and optimized benchmark results in the benchmark response:

- **Baseline (as-is):** no source-code modifications permitted. Library substitutions permitted if those libraries will be available at system delivery. Reasonable changes to compilation options permitted unless otherwise prohibited by the benchmark README.
- **Ported:** minimal source code changes required solely to enable execution on the proposed architecture are permitted. Examples include addition or modification of directives or pragmas, and/or replacement of existing architecture-specific language constructs with another well-documented language or interface (e.g., CUDA <-> HIP). Ported should not be reported without baseline, unless baseline is not possible.

- **Optimized:** additional code modifications allowed if they will be available in maintainable form at system delivery. For each benchmark, newer versions of the benchmark source code may be used if these versions are publicly available at the time of machine arrival. No surrogate models and no reduced-precision modifications are allowed unless permitted by the benchmark README.md

A baseline result is required whenever possible; a ported result may be provided in place of a baseline result if the baseline result is not possible; ported in addition to baseline is optional and optimized is fully optional. If any exception to these rules exists for a particular benchmark, it will be clearly stated in the corresponding benchmark README.

8.2 Application Benchmarks

Table 2 – ESIF-HPC-4 Application Benchmarks

Benchmark	Description
VASP*	Plane-wave electronic structure code
WRF	Weather simulation code (CPU-only)
Kynema-sgf	Computational fluid dynamics code
BerkeleyGW	Many-body perturbation theory
LAMMPS*	Molecular dynamics code
MLPerf-DeepCAM*	Deep learning training workload
Sienna	Energy systems analysis

*Included in application throughput improvement metric

8.3 Microbenchmarks

Table 3 – ESIF-HPC-4 Microbenchmarks

Benchmark	Description
OSU Microbenchmarks	MPI communication tests
STREAM	Memory bandwidth
HPL	Dense linear algebra
GPU-GPU collective	GPU collective communication

8.4 I/O Benchmarks

Table 4 – ESIF-HPC-4 I/O Benchmarks

Benchmark	Description
IOR	Parallel file system read/write performance
mdtest	Metadata performance
FIO	Single-node storage performance

8.5 Benchmark Requirements

8.5.1 Benchmark Execution Requirements

[TR-1] The Offeror shall provide performance results for all benchmark listed in Tables 2-4 using the prescribed input and instructions provided in the ESIF-HPC-4 GitHub repository.

8.5.2 Benchmark Documentation

[TR-1] The Offeror shall provide a complete benchmark report as an appendix to the Technical Volume and return the ESIF-HPC-4 Benchmark Results spreadsheet.

8.5.3 Allowed Modifications

[TR-1] The Offeror shall adhere to the definitions of baseline, ported, and optimized benchmark runs and any benchmark-specific constraint defined in the provided README.md files.

8.5.4 Application Throughput Improvement Metric

[TR-1] The Offeror shall compute and report the ESIF-HPC-4 application throughput improvement metric using the prescribed benchmark configurations, reflecting all system design constraints defined elsewhere in this document (including the CPU:GPU FLOPS ratio in Section 4.1.2), and shall demonstrate improvement over the provided reference results. Partial-node runs are permitted only when the benchmark workload cannot utilize at least half of a node's resources, and only if the resulting performance surpasses the provided Kestrel reference result.

8.5.5 Licensing for Benchmark Tools

[TR-2] The Offeror shall provide temporary licenses for any compiler, libraries, or profiling tools used to obtain benchmark performance results that are not part of the proposed delivered system.

8.5.6 Reproducibility

[TR-2] The Offeror shall describe the environment, build settings, runtime configuration, and system tuning parameters used to obtain reported benchmark results to enable NLR to reproduce performance.

8.5.7 Optional Optimizations

[TR-3] The Offeror may provide optimized benchmark results provided that all source-code changes will be made available to NLR in maintainable form at system delivery.

9.0 Facilities and Site Integration

NLR's HPC data center located in the Energy Systems Integration Facility (ESIF) is designed to be one of the most energy-efficient data centers in the world. Using best-in-class technologies, the data center serves as a showcase facility for demonstrating data center energy efficiency for AI and HPC facilities, in addition to providing computational production operations.

The ESIF HPC data center facility attributes are provided in Table 5 - ESIF-HPC-4 Facility Specifications. Diagrams showing the ESIF data center floor plan with proposed location of ESIF-HPC-4 systems along with a mechanical and electrical overview is in Figure 6: ESIF HPC Data Center Layout. These are intended to assist Offerors in understanding the integration requirements for ESIF-HPC-4 and related systems. These are subject to change prior to the ESIF-HPC-4 deployment.

9.1 Power and Cooling

The Offeror shall describe the power, cooling, mechanical and monitoring characteristics of the proposed system and how it integrates into the ESIF facility.

9.1.1 Environmental and Operational Conditions

9.1.1.1 Environmental Operating Limits

[MR] The Offeror will describe the environmental operating ranges of the system. Include water input, output and flow rates for water cooled equipment. Include the operating conditions for any air-cooled equipment including power draw, air input ranges and expected output to data center at 78°F air input.

The Offeror shall ensure that the proposed system operates reliability with the ESIF facility limits:

9.1.1.1.1 [TR-2] Liquid cooling supply temperatures up to 75°F (24°C)

9.1.1.1.2 [TR-2] Data center ambient air temperatures of 78°F +/- 2°F

9.1.1.2 System Power Characteristics

[MR] The Offeror will describe the power characteristics of the system, including total system power and per rack type power draw.

The Offeror shall provide an estimate of power for the RFP response and verify sustained power on the awarded system:

9.1.1.2.1 [TR-2] Idle system power

9.1.1.2.2 [TR-2] Sustained system power, defined as 75% of sustained load using HPL

9.1.1.2.3 [TR-2] Peak system power, defined as 100% of sustained load using HPL

9.1.2 Power Distribution

9.1.2.1 Supported Power Configurations

[TR-1] The Offeror shall describe all power requirements for compute, accelerate, and storage racks. The ESIF facility provides:

9.1.2.1.1 [TR-1] 3-phase 480V AC for racks not requiring N+1 redundancy, 480V AC or 240V AC for racks requiring N+1 (240V via APC PDPM288G6H, maximum 40 amp draw per circuit.)

9.1.2.1.2 [TR-2] If other voltages are required (e.g. 415V, 208V, 120V), the Offeror shall provide any necessary transformers.

9.1.2.2 Phase Balancing Requirements

[TR-2] If the system uses line-to-neutral power supplies, the Offeror shall describe required phase-balancing characteristics and verification methods.

9.1.2.3 Electrical Connections

[TR-2] NLR will provide all facility-side power cabling up to the ESIF-HPC-4 system boundary. The Offeror is responsible for providing all power cabling internal to the proposed system, including cable trays or containment integrated with and spanning between the system cabinetry. The Offeror shall list all electrical connections required at the rack level and describe cable routing, containment, and connector types.

9.1.2.4 Power Interruption Ride-Through

[TO-1] The Offeror may describe any built-in ability for nodes to remain operational at the OS Level during power interruptions up to 5 seconds (e.g. supercapacitor or in-rack battery technologies)

9.1.2.5 Local Rack-Level Disconnects

[TR-3] The Offeror may provide local rack-level electrical disconnect switches to simplify servicing components that are not hot-swappable. If provided, these disconnect mechanisms should integrate cleanly with the ESIF power distribution configuration and support safe isolation of individual racks during maintenance activities.

9.1.3 Cooling Integrations

9.1.3.1 Liquid Cooling Integration

[TR-1] The Offeror shall describe the liquid cooling requirements of the system including:

- Supply/return flow rates
- Pressure ranges
- Materials in contact with coolant
- Required connection points
- Required intermediary cooling distribution unit (if applicable)

9.1.3.2 Facility Return Temperature

[TR-1] To support energy-recovery operations, please verify the system shall be capable of delivering return water temperatures of $\geq 95^{\circ}\text{F}$ (35°C) under normal load.

9.1.3.3 Air Cooling

The ESIF HPC Data Center was designed primarily for liquid cooling. There is a modest 250KW of facility provided air cooling.

9.1.3.3.1 [TR-2] The Offeror shall describe any air-cooled subsystems and confirm that the proposed solution does not require more than 200kW

9.1.3.3.2 [TR-3] If the proposed solution exceeds 250KW utilization of in-row coolers, rear door heat exchangers are acceptable to maintain room neutrality. Provide details on the number, type and connection requirements to the ESIF facility.

9.1.3.4. Cooling Capacity Across Load States

The Offeror shall provide expected supply and return water temperatures at:

9.1.3.4.1 [TR-2] Idle

9.1.3.4.2 [TR-2] Sustained system power, defined as 75% of sustained load using HPL

9.1.3.4.3 [TR-2] Peak system power, defined as 100% of sustained load using HPL

And shall describe how these values may be monitored continuously.

9.1.3.5 Intermediary Cooling Distribution Unit (CDU) Responsibility

[TR-1] If the proposed system's cooling infrastructure requires pressure, flow, temperature, or water-chemistry characteristics that differ from those provided by the ESIF Energy Recovery Water (ERW) system, the Offeror shall provide, install, and integrate any required intermediary cooling distribution unit (CDU) or heat-exchange subsystem. The Offeror is responsible for ensuring full compatibility between the system's secondary cooling loop and the facility ERW primary loop.

9.1.4 Mechanical and Facility Integration

9.1.4.1 Wetted Materials Disclosure

[TR-1] The Offeror shall provide a complete list of all wetted materials used in the secondary cooling loop and support chemical approval prior to shipment.

9.1.4.2 Mechanical Interfaces

The Offeror shall describe:

9.1.4.2.1 [TR-1] Rack manifold layouts

9.1.4.2.2 [TR-1] Hose and fittings

9.1.4.2.3 [TR-1] Distribution piping or manifolding integrated with each rack

9.1.4.2.4 [TR-1] Any required facility-side modifications

9.1.4.3 Rack Labeling

[TR-2] All racks, nodes, switches, and enclosures shall be clearly labeled with durable unique identifiers visible from front and rear, including node-serial identifiers matching administrative designations.

9.1.4.4 Remote Environmental Monitoring

[TR-2] The Offeror shall describe all available sensors and telemetry interfaces supporting facility integration, including coolant temperatures/flows, rack-level thermal sensors, and relevant power/cooling metrics exportable into ESIF monitoring systems.

9.2 Delivery and Installation

The Offeror shall describe the delivery, installation, and integration approach for ESIF-HPC-4, including coordination with NLR personnel, on-site activities, and system bring-up procedures. The following information and requirements apply to all proposed systems.

9.2.1 Delivery

NLR will provide personnel to supervise unpacking and movement of equipment within the facility. Offerors should refer to Table 5: ESIF-HPC-4 Facility Specifications for shipment dimensions, weight limits, and access constraints. All personnel performing on-site work must meet NLR facility access requirements and complete a mandatory safety briefing before beginning work.

Shipping containers should be clearly labeled with the subcontract number, Offeror name, serial numbers, and content descriptions. For multi-rack systems, each rack and its shipping container shall be clearly labeled to support orderly placement and assembly in the ESIF facility.

[TR-2] The Offeror shall coordinate with NLR prior to delivery, providing shipment schedules, expected delivery configurations (e.g. assembled vs partially assembled racks), and any special handling instructions.

9.2.2 Installation

Prior to delivery, the Offeror shall hold coordination meetings with NLR Advanced Computing Systems and Operations staff, conduct a site visit, and participate in a joint facility readiness review. The Offeror should communicate all hardware, software, environmental, and facility integration requirements necessary to support immediate installation upon delivery.

9.2.2.1 Installation Responsibilities

[TR-1] The Offeror shall install, interconnect, and configure all components of the ESIF-HPC-4 system, including compute, storage, networking, power, cooling, and management subsystems, in accordance with ESIF facility constraints.

9.2.2.2 System Initialization and Integration

[TR-2] The Offeror shall perform initial power-on, diagnostics, and software initiation processes, and shall support NLR in integrating the system into facility power, cooling, monitoring, and network environments.

9.2.2.3 On-Site Safety and Work Practices

[TR-2] The Offeror shall ensure that all on-site personnel follow NLR safety practices related to electrical work, mechanical handling, equipment lifting, and ladder use.

10.0 Maintenance and Support

NLR seeks a maintenance and support strategy that maximizes availability, minimizes unplanned downtime, and ensures continued operational readiness of the ESIF-HPC-4 system. The proposed maintenance model shall support DOE's quarterly reporting requirements and NLR's current availability target of 95% across the federal fiscal year.

10.1 Maintenance and Support

10.1.1 Multi-year support period

[TR-1] The Offeror shall propose technical services, warranty, maintenance, and support for a minimum of five (5) years following system Acceptance.

10.1.2 Hardware Maintenance and Replacement

The Offeror shall describe the hardware support model, including:

10.1.2.1 [TR-1] Expected replacement time for failed Field Replaceable Units (FRUs) and Customer Replaceable Units (CRUs)

10.1.2.2 [TR-1] Ability to perform hot-swap or warm-swap replacement where supported by system architecture

10.1.2.3 [TR-1] Processes for replacing components exhibiting pre-failure symptoms (e.g., excessive correctable errors)

10.1.2.4 [TR-1] No component shall remain out of service for more than seven (7) calendar days unless agreed upon by NLR.

10.1.3 On-site Spares

[TR-1] The Offeror shall provide a recommended on-site spares cache sized to sustain normal repair actions for at least four (4) weeks without replenishment. The Offeror shall replenish consumed spares within two (2) weeks, unless otherwise agree.

The Offeror shall propose an alternative support model only if it provides equal or better availability outcomes.

10.1.4 Spares Inventory Management

[TR-2] It is desirable that the Offeror provide an inventory management capability that supports on-demand reporting of available, consumed, and back-ordered parts, including part identifiers and firmware revisions.

10.1.5 Software and Firmware Updates

The Offeror shall provide all required software updates and firmware updates for the duration of the maintenance period, including:

10.1.5.1 [TR-1] Security and cybersecurity patches

10.1.5.2 [TR-1] Vendor-provided functionality updates

10.1.5.3 [TR-1] Support for community software components included with the system

10.1.6 Support Access and Escalation

The Offeror shall describe:

10.1.6.1 [TR-1] Supported communication channels (e.g. web portal, phone, email, on-site support)

10.1.6.2 [TR-1] Issue tracking processes

10.1.6.3 [TR-1] Escalation paths and expected response time for severity levels

10.2 Resilience, Reliability, and Availability Metrics

10.2.1 Availability Target

[TR-1] The Offeror shall propose a system availability target consistent with NLR's operational goals. Available shall be computed as:

$$\text{Availability} = \frac{\text{time in period} - \text{unscheduled downtime}}{\text{time in period} - \text{scheduled downtime}} \times 100$$

10.2.2 Mean Time Between Interrupt (MTBI)

[TR-2] The Offeror shall propose a Mean Time Between Interrupt (MTBI) for the system and describe how this metric is calculated.

10.2.3 System Power-on and Shutdown Sequencing

[TR-2] The Offeror shall describe the full system power-on and power-off sequences, including rack- and component-level dependencies and estimated elapsed times.

10.3 Media Retention

10.3.1 Data Storage Media Handling

[TR-1] NLR retains all non-volatile media on-site at the end of service life. The Offeror shall provide a process for ensuring NLR can remove or retain such media and shall describe any component-level constraints. NLR will perform destruction or sanitization in accordance with NLR policy.

11.0 Project Management

The Offeror shall describe a project management approach that ensures timely delivery, effective risk management, and successful integration of the ESIF-HPC-4 system. The project management framework shall support close technical collaboration between the Offeror and NLR throughout system design, delivery, installation and acceptance.

11.1 Project Management and Organization

11.1.1 Project Management Approach and Organization

[TR-1] The Offeror shall describe its overall project management approach and proposed project organization, including key roles, responsibilities and escalation paths. At a minimum, this shall include a Project Manager, a Technical Lead or System Architect, and an executing-level point of contact.

11.1.2 Technical Decision Point

[TR-1] The Offeror and NLR shall conduct a Technical Decision Point review approximately 6 months prior to system delivery to confirm the final system configuration, reassess performance targets, and account for technology readiness.

11.2 Project Planning, Schedule, and Risk

11.2.1 Project Schedule and Milestones

[TR-1] The Offeror shall describe the proposed project schedule from contract award through system acceptance, including major milestones and dependencies. In the Price Proposal, the Offeror shall propose payment milestones tied to objective delivery or acceptance-related events.

11.2.2 Risk Identification and Mitigation

[TR-1] The Offeror shall identify key technical, schedule, and supply-chain risks and describe mitigation strategies for each.

11.3 Technical Coordination and Acceptance

11.3.1 Project Kick-Off and Ongoing Coordination

[TR-2] The Offeror shall participate in a project kick-off meeting following contract award and in regular technical coordination meetings with NLR throughout the project lifecycle.

11.3.2 Acceptance Testing

[TR-1] The Offeror shall support acceptance testing of the ESIF-HPC-4 system, including functionality, performance and stability testing. Acceptance testing requirements shall be derived from the Technical Specification, and the detailed acceptance test plan shall be finalized during contract award negotiations.

12.0 Training and Documentation

The Offeror shall describe training and documentation sufficient to enable effective administration and use of the ESIF-HPC-4 system by NLR staff and users throughout the system lifecycle.

12.1 System Administration Training

12.1.1 Initial Systems Training

[TR-1] The Offeror shall provide initial systems training targets to NLR operations and applications support staff to enable effective administration, operation, and support of the ESIF-HPC-4 system. This training shall be completed prior to system delivery or initial operational readiness.

12.1.2 Ongoing and Update Training

[TR-2] The Offeror shall describe provisions for additional training in the event of significant changes to system hardware, software, or administrative procedures, including changes introduced through upgrades or early-access systems.

12.2 End User Training

12.2.1 Initial End User Training

[TR-2] The Offeror shall provide initial end user training following system acceptance to support effective use of the ESIF-HPC-4 system, including guidance on programming environments, system architecture, and efficient use of system resources (e.g., how users can control process and memory placement, binding, and pinning for performance). Additional end user training may be requested by NLR in the event of significant architectural or software changes introduced during the system lifecycle.

12.2.2 Training Materials and Recording Rights

[TR-2] The Offeror shall describe how NLR is granted the right to record, reproduce, and distribute end-user training materials (including audio, video, and written content) for internal use by ESIF-HPC-4 users.

12.3 Documentation

12.3.1 Documentation Scope and Availability

[TR-1] The Offeror shall provide documentation sufficient to support system installation, administration, operation, and user access to ESIF-HPC-4. Documentation shall cover system hardware, system software, management tools, user environments, and major subsystems.

12.3.2 Documentation Access and Rights

[TR-1] The Offeror shall describe how documentation is delivered to NLR and shall grant NLR the right to internally publish, distribute, and use the documentation for the duration of system operation.

12.3.3 Documentation Maintenance

[TR-2] The Offeror shall describe how documentation is maintained and updated over time to reflect changes to system hardware, software, or configuration in a timely manner.

Appendix A: Benchmarking Application Throughput Improvement Metric

We define the application throughput improvement metric as the weighted geometric mean of the ratios of throughput on the offered machine and reference machine for each application. We define application-level throughput as the harmonic mean of the rate of job completion on each relevant hardware type, where the rate of job completion on a given hardware type is defined as:

$$R_{app} = \frac{N_{nodes}^{total}}{N_{nodes}^{job}} \frac{1}{t}$$

Where N_{nodes}^{total} denotes the total number of nodes of a given type (e.g., CPU-only or accelerated) on the machine, N_{nodes}^{job} denotes the number of nodes of the given type that the application was run on, and t denotes the performance result for the application (e.g., runtime of the application). $\frac{N_{nodes}^{total}}{N_{nodes}^{job}}$ then represents the number of copies of the benchmark that can be executed simultaneously, so that R_{app} is the theoretical number of jobs per second that could be completed for the application on the given hardware type. The throughput of the application on the machine is given as:

$$R_{app}^{machine} = \frac{1}{\frac{p_{CPU}}{R_{app}^{CPU}} + \frac{p_{GPU}}{R_{app}^{GPU}}}$$

Where p represents the portion of jobs completed on each side of the machine, $p_{CPU} + p_{GPU} = 1$. We compute the application-level throughput as a harmonic mean because, on the reference system, many users execute these applications on both CPU-only and accelerated nodes, and thus providing a reference throughput for only one portion generates a skewed result. However, on the next-generation system, we anticipate that users will typically not execute these applications on mixed hardware. While NLR will provide p_{CPU} and p_{GPU} for each application, the Offeror is free to modify these weights, and in cases where $p_{CPU} = 1$ or $p_{GPU} = 1$, may optionally provide results on both hardware types and update p_{CPU} and p_{GPU} to maximize the application throughput improvement metric, if desired.

Finally, the throughput metric is defined as:

$$Throughput\ Improvement = \prod_{i=1}^{M_{apps}} \left(\frac{R_i^{offered}}{R_i^{ref}} \right)^{w_i}$$

Where w_i denotes the weight for each application throughput improvement in the weighted geometric mean, $0 < w_i < 1$, $\sum w_i = 1$. These weights will be provided by NLR and cannot be modified.

Appendix B: Facilities & Site Integration

Table 5: ESIF-HPC-4 Facility Specifications

Physical Information	Location	NLR – HPC Data Center located in the Energy Systems Integration Facility (ESIF), Golden, Colorado
	Altitude	Altitude 5,800 feet above mean sea level.
	Seismic	n/a
	Floor	17-ft raised floor; Data center floor consists of 2ft x 2ft floor tiles at one foot above steel beam support structure below – there is a safety mesh 2-inches below floor tiles.
	Floor Loading	Floor loading at an individual rack location shall not exceed 900 pounds per square foot. The entire data center floor is rated for an average load of 250 pounds per square foot. Equipment aisle spacing is normally 6 feet but should be increased if necessary to maintain an average load of 250 pounds per square foot in the portion of the datacenter where the system will be installed. Significantly higher loads at rack locations and minimal loads in equipment aisles and other access areas are permitted; floor loading shall not exceed concentrated load of 2,500 pounds on one square inch.
	Ceiling	11-ft ceiling; 18-inches required for sprinkler head clearance.
	Shipment Dimensions and Weight	All doors and pathways are 6-ft in width and 7-ft 11-in in height, or larger; NLR will want to review the delivery mechanism for any system component over 6,000 pounds
Environmental, Mechanical	Water Cooling	The system must operate within ASHRAE TC 9.9 temperature ranges; Water supplied by the facility ERW loop will vary between 60°F and 75°F (this temperature has typically been held between 65°F - 70°F); Pressure drop across HPC data center and associated control valves shall be kept below 20 PSI; The return pressure of the facility ERW loop is typically maintained to around 30 PSI.
	Air Cooling	The system must operate within ASHRAE TC 9.9 temperature ranges; Space temperatures within the data center are 78°F +/-2°F, with relative humidity not more than 60% (a make-up air unit attempts to keep relative humidity above 20% in the data center); Up to 200 kW of ESIF-HPC-4 equipment may be air-cooled using complete hot-aisle containment or separation.

Electrical	Power Availability	Power Availability	
		Line	UPS
		7MW max	300kW max
		Panel info	
		Line 3x 480VAC 800 amp	UPS fed from APC modular PDU's (See section 5.1)
	Power Cabling	Facility power to racks located in ESIF-HPC-4 area can be fed from above or below floor.	
Network	Cabling	Shall be provided by Offeror with cable trays or cable containment integrated with and spanning between the system cabinetry.	

Data Center Floor Layout

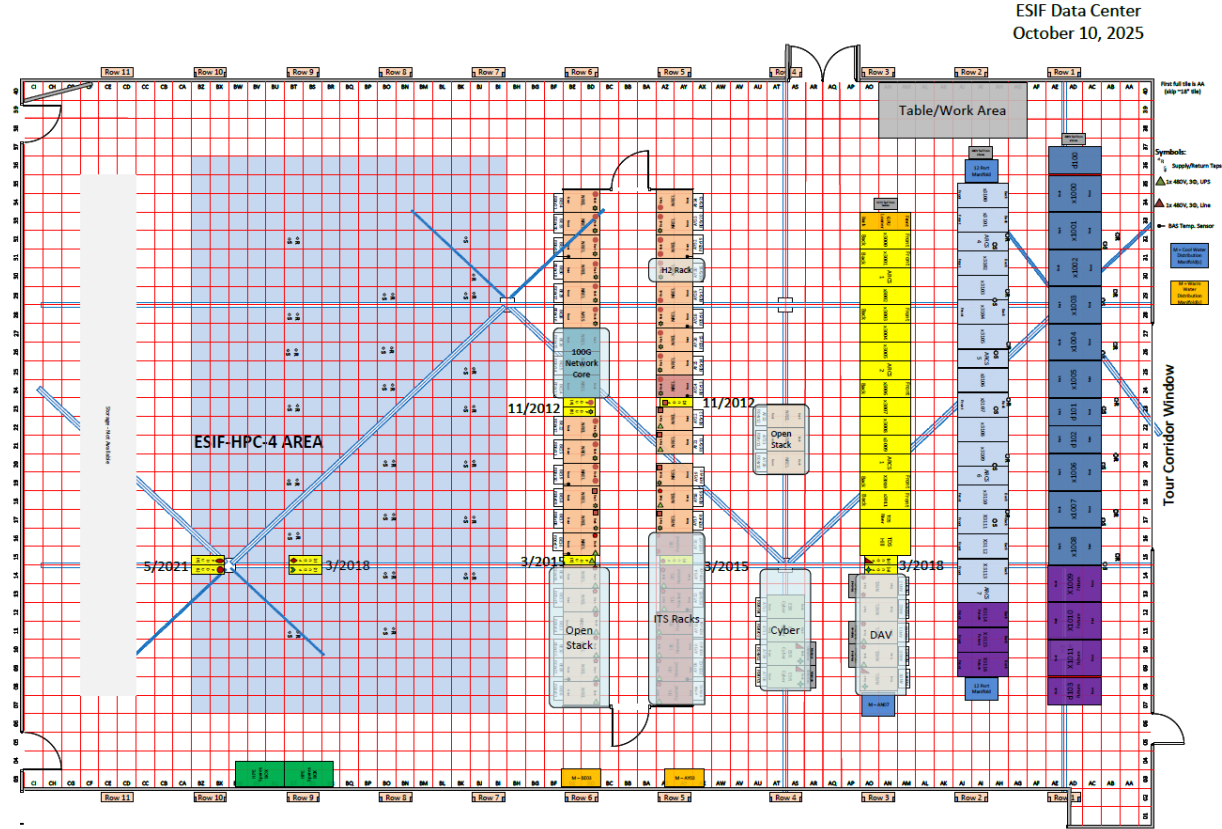


Figure 6: ESIF Data Center Floor Layout. ESIF-HPC-4 Area denotes location where majority of the ESIF-HPC-4 system will reside.